

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ROTATIONAL MOTION

LECTURE-2

PHYSICS

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Topics To Be Covered

- 1 Revision ✓
- 2 Motion of COM
- 3 Rotational Motion
- 4 Angular Distance/Angular Velocity/Angular Acceleration
- 5 Angular Momentum ✓

Topics To Be Covered

- 6** Direction of Angular Momentum ✓
- 7** Torque ✓
- 8** Direction of Torque ✓

“Mehnat

Zarori hai”!

Papa → Auto

$$\frac{497}{500}$$

SA	B
F	K

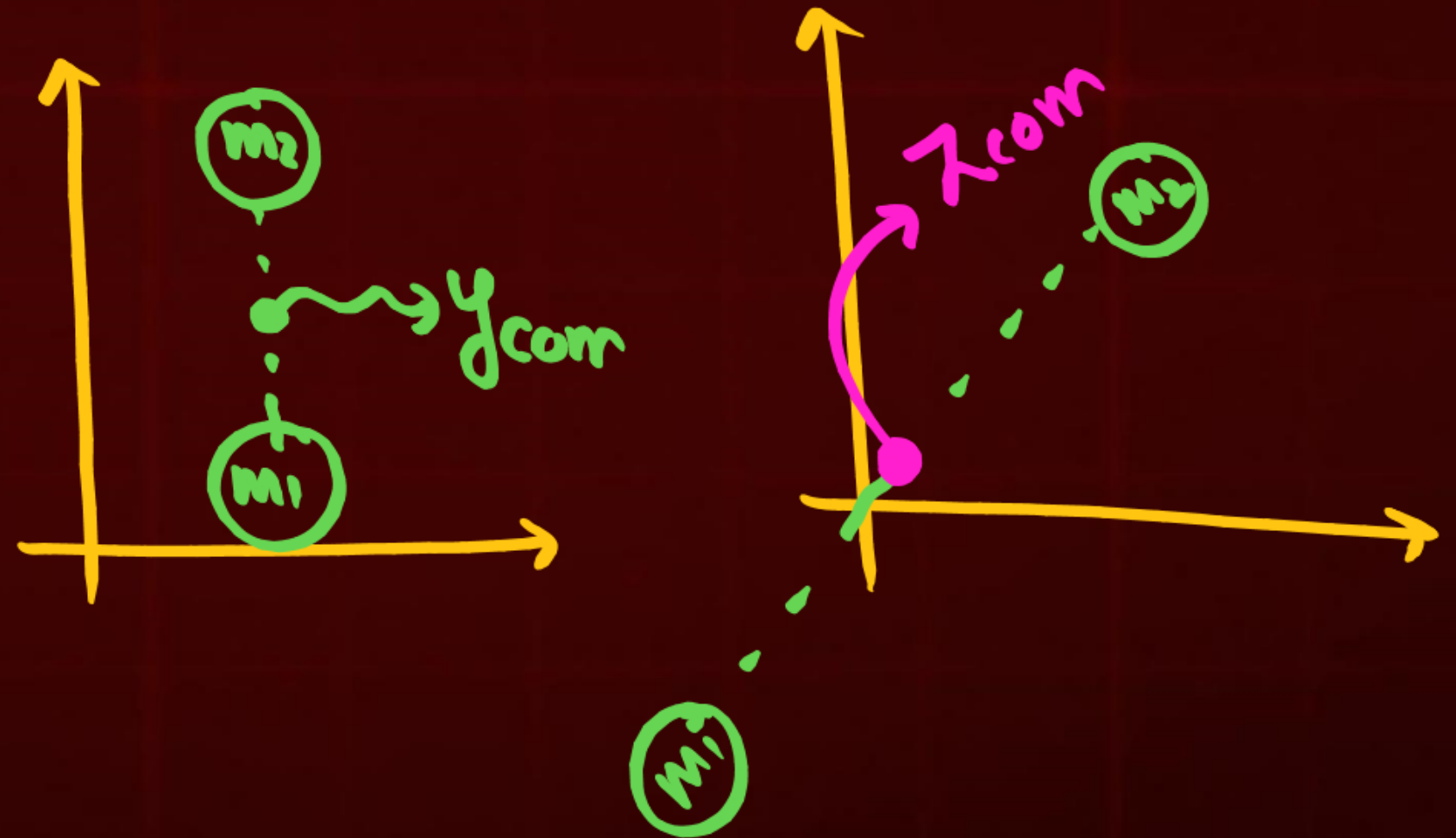
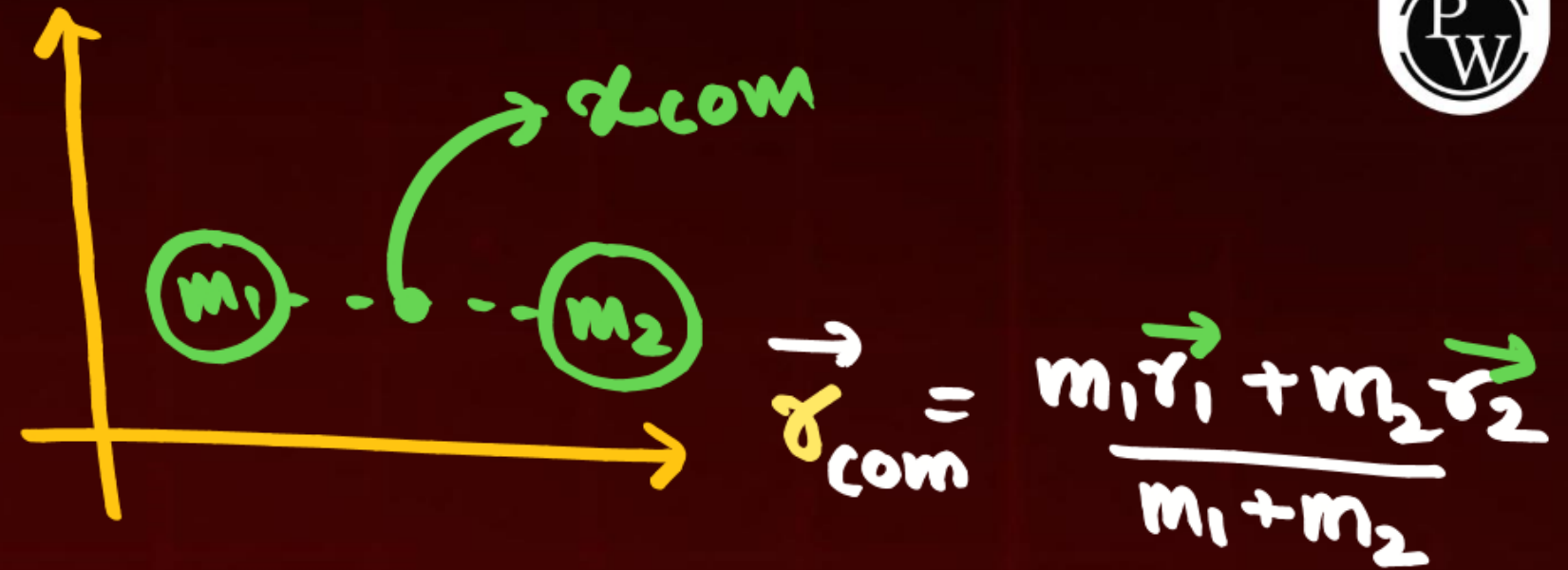
Recap From 1st Lecture

Position of Com

$$x_{\text{com}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

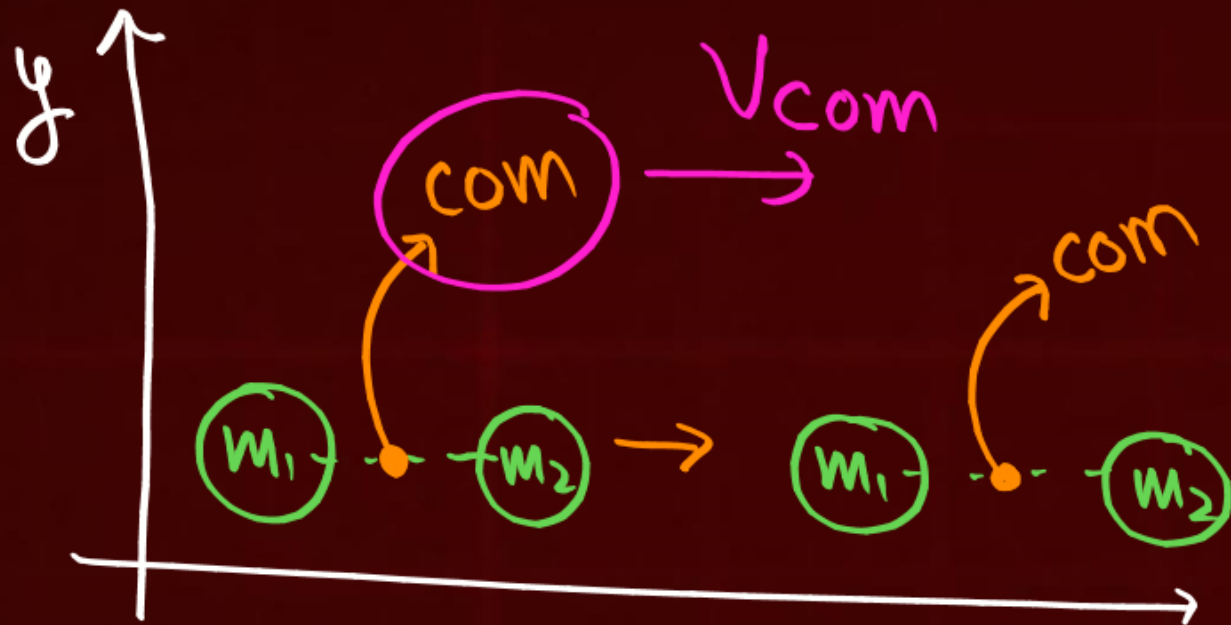
$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$z_{\text{com}} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$

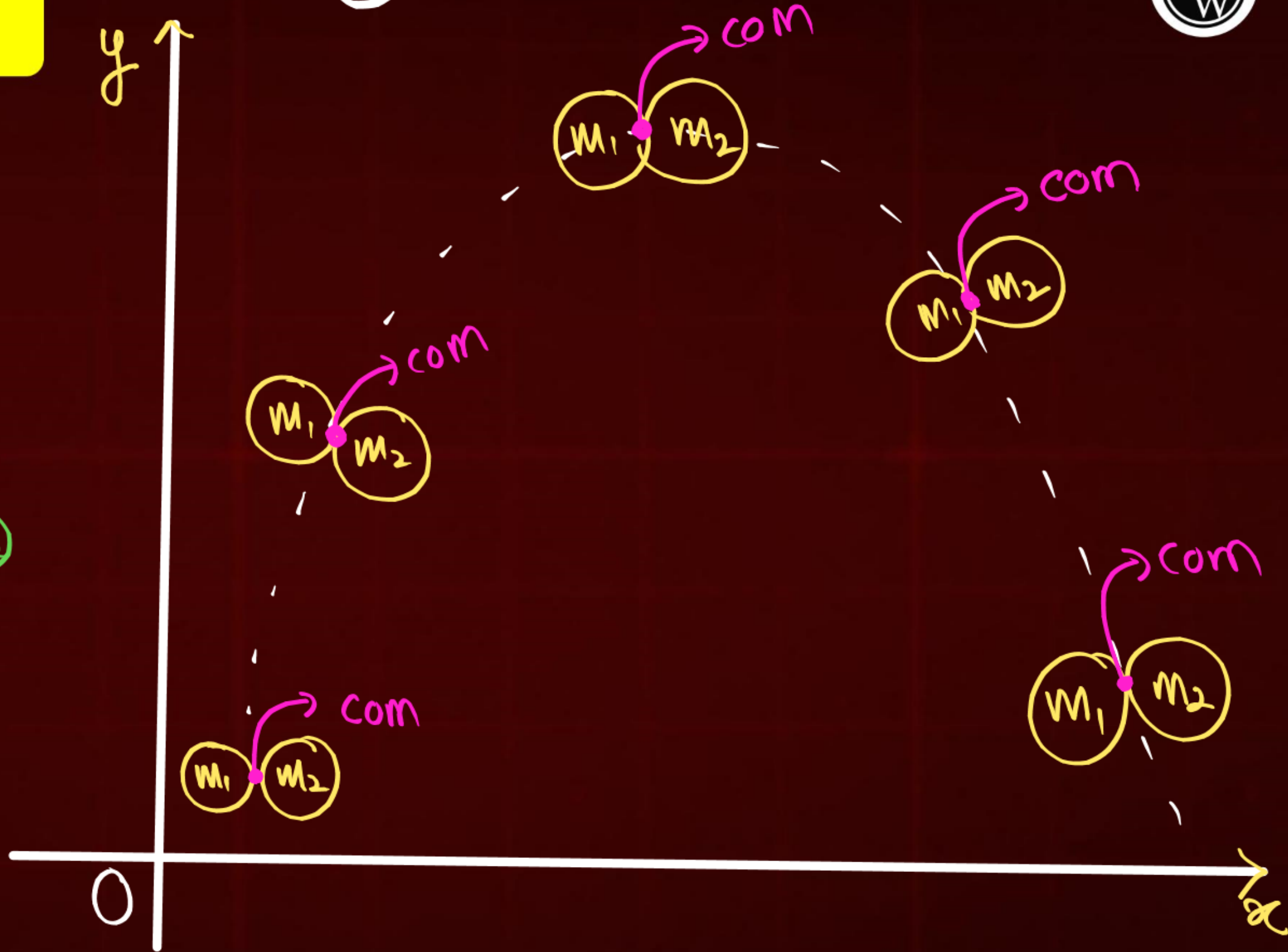


MOTION OF COM

When the COM moves



lecture 2



Velocity of COM ($\vec{V}_{\text{com}}$)

Velocity is Rate of change of displacement

$$\# \quad v = \frac{d\vec{r}}{dt}$$

$$\# \quad \vec{r}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$\vec{r}_{\text{com}}(m_1 + m_2) = m_1 \vec{r}_1 + m_2 \vec{r}_2$$

General formula of Velocity of com

On differentiating b/s

$$\frac{d\vec{r}_{\text{com}}(m_1 + m_2)}{dt} = \frac{d}{dt}(m_1 \vec{r}_1 + m_2 \vec{r}_2)$$

$$\vec{V}_{\text{com}}(m_1 + m_2) = \frac{d}{dt} m_1 \vec{r}_1 + \frac{d}{dt} m_2 \vec{r}_2$$

$$\vec{V}_{\text{com}}(m_1 + m_2) = m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt}$$

$$\vec{V}_{\text{com}}(m_1 + m_2) = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

$$\vec{V}_{\text{com}} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{(m_1 + m_2)} \quad \text{--- (1)}$$

$$\vec{V}_{\text{com}} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

General
formula.

x

$$\vec{V}_{x\text{com}} = \frac{m_1 \vec{v}_{x1} + m_2 \vec{v}_{x2}}{m_1 + m_2}$$

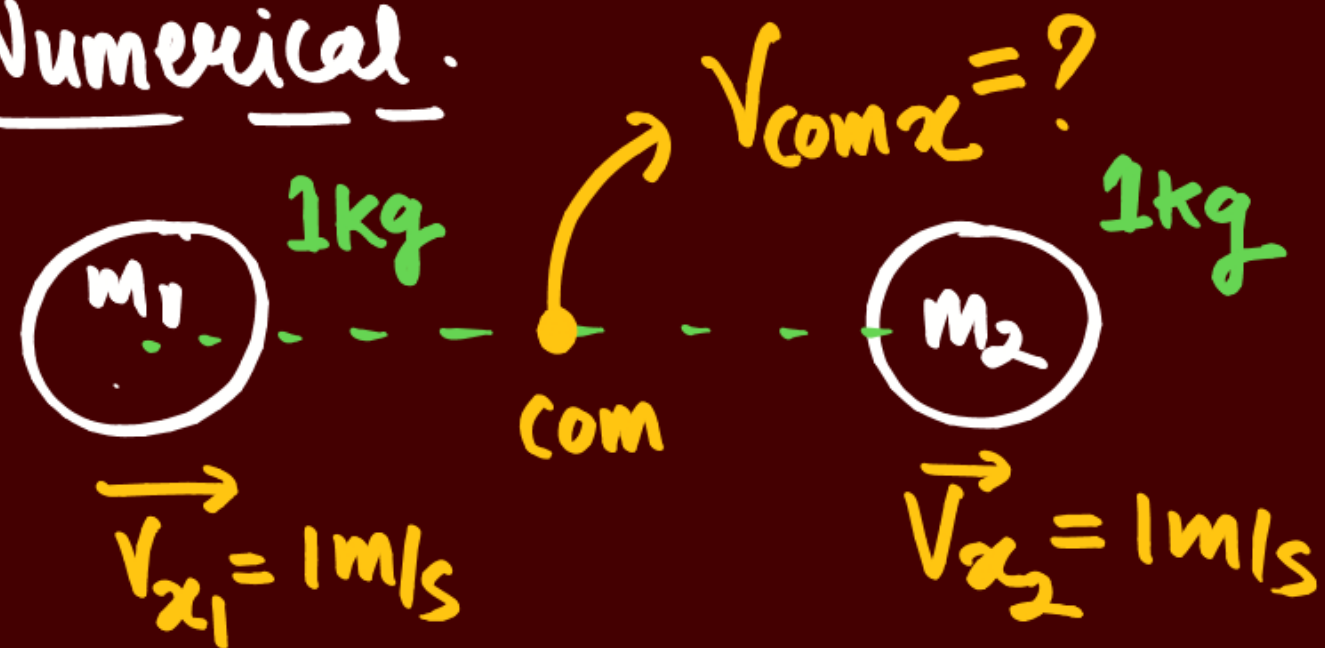
y

$$\vec{V}_{y\text{com}} = \frac{m_1 \vec{v}_{y1} + m_2 \vec{v}_{y2}}{m_1 + m_2}$$

z

$$\vec{V}_{z\text{com}} = \frac{m_1 \vec{v}_{z1} + m_2 \vec{v}_{z2}}{m_1 + m_2}$$

Numerical.



$$V_{x\text{com}} = ?$$

Sol

$$V_{x\text{com}} = \frac{m_1 \vec{v}_{x1} + m_2 \vec{v}_{x2}}{m_1 + m_2}$$
$$= \frac{1 \times 1 + 1 \times 1}{1 + 1} = \frac{1 + 1}{2} = \frac{2}{2} = 1\text{m/s}$$

Acceleration of COM

- The rate of change of velocity

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{V}_{\text{com}} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

$$\vec{a}_{\text{com}} \quad \vec{V}_{\text{com}}(m_1 + m_2) = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

on differentiating both

$$\frac{d}{dt} \vec{V}_{\text{com}}(m_1 + m_2) = \frac{d}{dt} (m_1 \vec{v}_1 + m_2 \vec{v}_2)$$

$$\vec{a}_{\text{com}}(m_1 + m_2) = \frac{d}{dt} (m_1 \vec{v}_1) + \frac{d}{dt} m_2 \vec{v}_2$$

$$\vec{a}_{\text{com}}(m_1 + m_2) = m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt}$$

$$\vec{a}_{\text{com}}(m_1 + m_2) = m_1 a_1 + m_2 a_2$$

$$\vec{a}_{\text{com}}(m_1+m_2) = m_1\vec{a}_1 + m_2\vec{a}_2$$

$$\vec{a}_{\text{com}} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1+m_2}$$

General formula of acc. of com

$$\vec{a}_{com} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2}$$

x

$$\vec{a}_{xcom} = \frac{m_1 \vec{a}_{x1} + m_2 \vec{a}_{x2}}{m_1 + m_2}$$

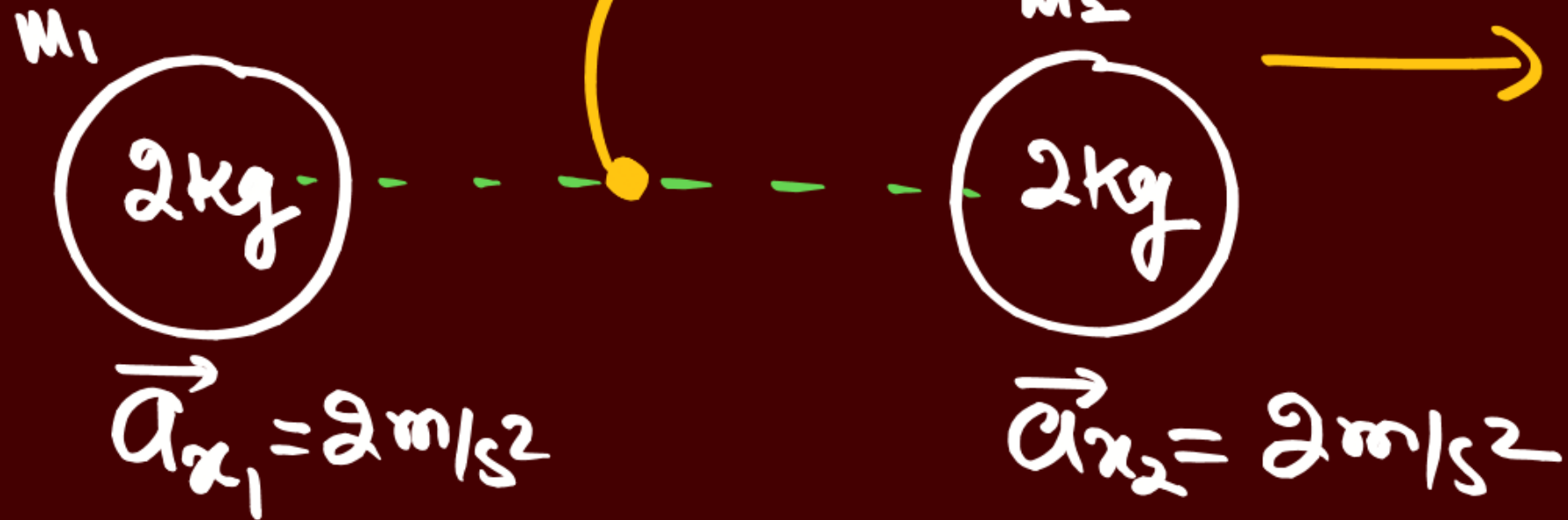
y

$$\vec{a}_{ycom} = \frac{m_1 \vec{a}_{y1} + m_2 \vec{a}_{y2}}{m_1 + m_2}$$

z

$$\vec{a}_{zcom} = \frac{m_1 \vec{a}_{z1} + m_2 \vec{a}_{z2}}{m_1 + m_2}$$

Q Numerical.



Sol

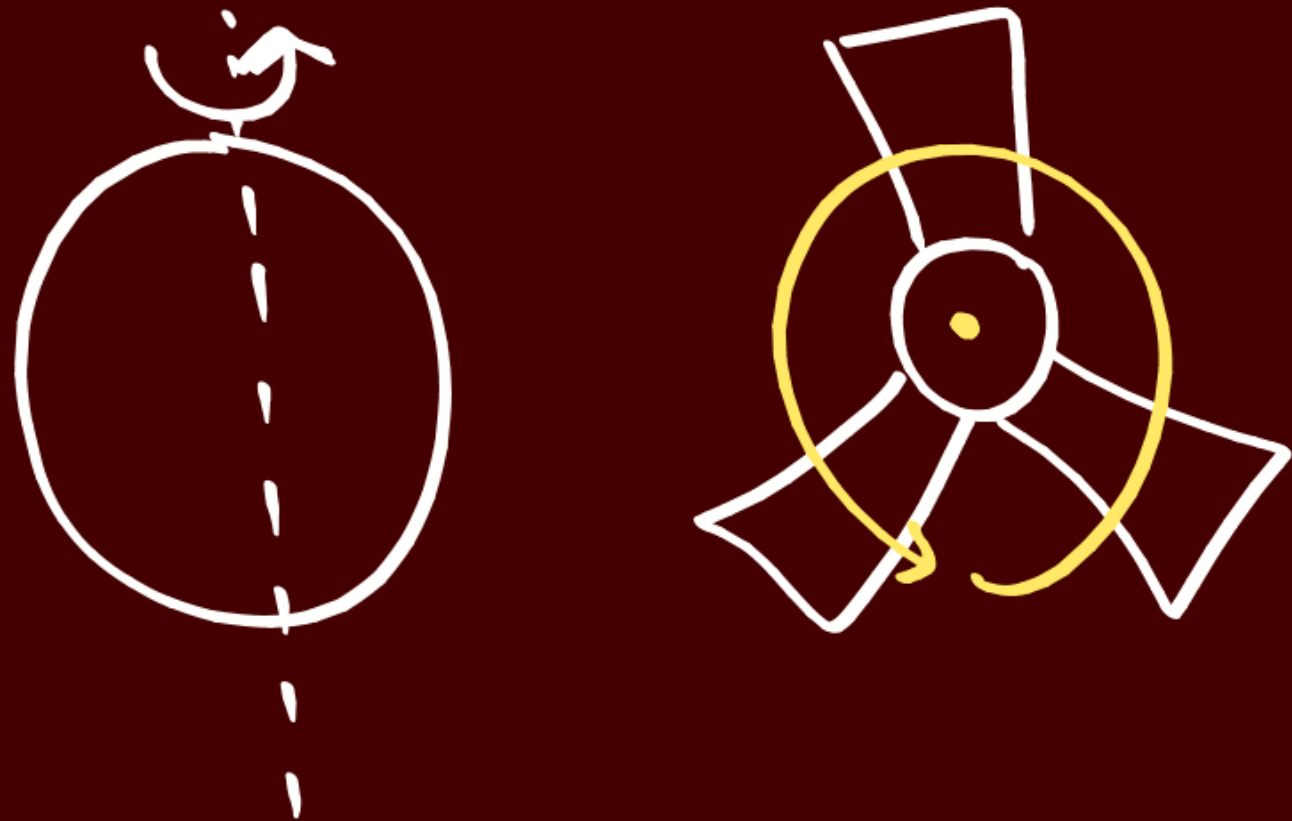
$$\vec{a}_{x,com} = \frac{m_1 \vec{a}_{x_1} + m_2 \vec{a}_{x_2}}{m_1 + m_2}$$
$$= \frac{2 \times 2 + 2 \times 2}{2 + 2} = \frac{4 + 4}{4} = \frac{8}{4} = \underline{\underline{2\text{m/s}^2}}$$

Rotational Motion

- When the Body Rotate about its own axis

Such a motion is

known as **Rotational Motion**



Angular displacement θ

Angular velocity ω

Angular acceleration α

① Angular displacement (θ)

the Angle Rotated
By the Body
During Rotation
Motion

$$\text{Angle} = \frac{\text{Arc}}{R}$$

$$\theta \rightarrow \begin{matrix} \text{degree} \\ \text{Radian} \end{matrix} \quad \theta = \frac{x}{R}$$

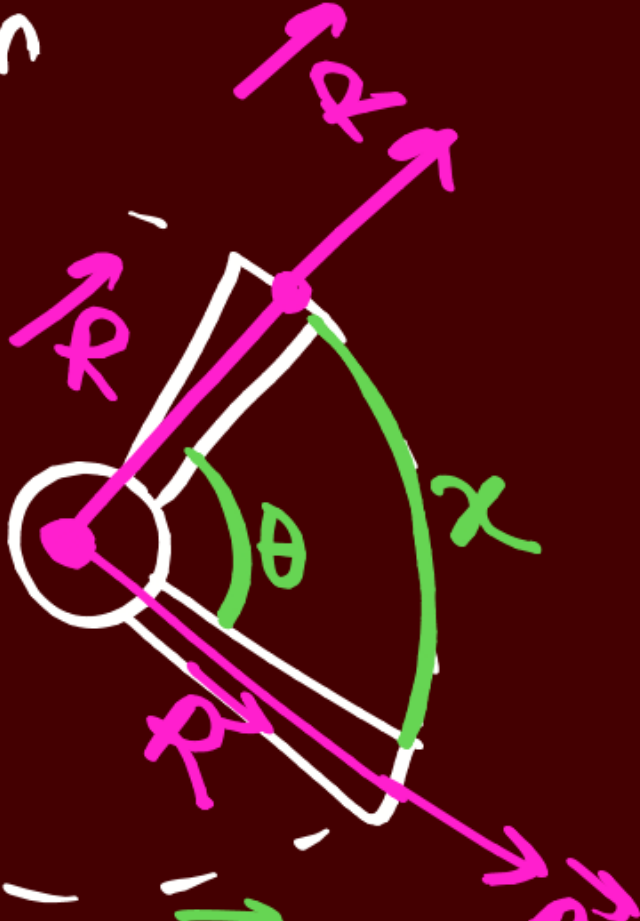
$\downarrow$
S.I.
unit

$$R\theta = x$$

$$x = R\theta$$

Displacement

Angular displacement.



Direction of $\vec{R}$ is from Centre to outside

② Angular velocity (ω)

Rate of change of
Angular displacement $\rightarrow$ S.I

$$\omega = \frac{\theta}{t} = \frac{\text{rad}}{\text{Sec}}$$

Neet

ω_0
Initial
Angular velocity

ω
Final
angular
velocity

$$\omega = \frac{\theta}{t}$$

$$\theta = \frac{x}{R}$$

$$\omega = \frac{x/R}{t}$$

$$\omega = \frac{x}{R} \times \frac{1}{t}$$

$$\omega = \frac{x}{Rt}$$

$$\omega = \frac{v}{R}$$

$$R\omega = v$$

$$v = R\omega$$

velocity

Angular
velocity

③ Angular Acceleration (α)

Rate of change of
Angular velocity (ω)

$$\alpha = \frac{\omega}{t}$$

unit

$$\alpha = \frac{\text{rad/sec}}{\text{sec}}$$

$$= \frac{\text{rad}}{\text{sec}^2}$$

$$\# \alpha = \frac{\omega}{t}$$

$$v = r\omega$$
$$\frac{v}{r} = \omega$$

$$\alpha = \frac{\frac{v}{r}}{t}$$

$$\alpha = \frac{v}{r} \times \frac{1}{t}$$

$$\alpha = \frac{v}{r \times t} a$$

$$\alpha = \frac{a}{r}$$

$$r\alpha = a$$

$$a = r\alpha$$

acceleration

angular acc.

$$x = R\theta$$

$$v = R\omega$$

$$a = R\alpha$$

.

SUMMARY



V_{com}

a_{com}

θ

ω

α

HOMework



Vector & Cross product.

Q

What is cross product of two
vector

“BELIEVE IN YOURSELF”

**“EVEN WHEN NO ONE
ELSE DOES.”**

Thank
You

